PIYUSH KUMAR

Software Engineer

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Summary

Computer Science undergraduate who builds and debugs production backend and AI systems through root-cause profiling and redesign; fixed a critical authentication flaw in a system in active use.

Experience

Software Intern, *AventIQ (Mittal Software Labs Ltd)* 07/2026 – Present
Delhi

- Developed the backend of an AI-driven LinkedIn campaign platform that runs campaigns end to end, from idea generation to publishing posts with image, video, GIF and PDF content; users only approve scheduled posts.
- Designed and implemented the background job system, post scheduler and multi-step AI content workflow that generate and publish posts autonomously, using Node.js, Express and MongoDB.
- Identified and fixed a critical authentication bypass in the production employee-management portal, closing an unauthorized-access path affecting all user accounts.
- Cut portal response time from 8–9 s to ~500 ms (~94%) by restoring a non-functioning Redis caching layer and rewriting the queries behind the slowest endpoints.

Frontend Developer Intern, *The Angaar Batch* 04/2026 – 05/2026
Remote

- Delivered and improved responsive interfaces and pages for the web portal of an edtech platform serving live production traffic.

Frontend Development Intern, *Verso (Wandrai)* 10/2025 – 11/2025
Delhi

- Owned the MVP frontend from scratch, including the landing page, auth flow and core application screens.
- Resolved dependency conflicts blocking integration with the existing codebase by migrating the build from Expo to React + Vite.
- Worked with the backend team on API integrations essential to the MVP launch.

Education

B.Tech, Computer Science, *KCC Institute of Technology & Management* 2023 – 2027
Greater Noida
CGPA: 7.87 / 10 (till 6th semester) | Class XII: 76.86% | Class X: 82% | Coursework: Data Structures, Algorithms, Operating Systems, Computer Networks, Database Systems, Probability & Statistics

Technical Skills

Languages
Python, C, C++, JavaScript, SQL

AI/ML
YOLOv8, OpenCV, Scikit-learn, NLP, LLM APIs, LLM security

Backend & Systems
FastAPI, Flask, MERN, REST APIs, background jobs and scheduling, Redis caching, query optimization, AWS (Lambda, Secrets Manager), Linux, Docker

Tools
Git, GitHub, Postman, VS Code, Jupyter, Hugging Face Spaces, Netlify

Projects

IronGuard — LLM Security Gateway, *Python · FastAPI · Vector DB · NLP*

- Engineered a multi-layer gateway that inspects LLM requests in real time for prompt injection, jailbreaks and PII leaks.
- Reduced end-to-end latency from 13,000 ms to 372 ms (~97%) by restructuring a serial inspection pipeline into parallel stages and removing redundant blocking calls, isolating each bottleneck through staged timing.
- Raised detection accuracy from 64% to 84.8% and true-positive rate from 60% to 89.9% on a 2,700-case adversarial test suite.
- Architected a hybrid scoring engine that runs a deterministic rule-based fast path ahead of the expensive semantic layer.

VisionX — Safety-Critical Object Detection, *Python · YOLOv8 · OpenCV*
Live demo: https://piyushkarn-visionx.hf.space/ Hackwave 2025, 1st place (Duality AI Track)

- Trained a YOLOv8s detector on synthetic space-station data to identify fire extinguishers, toolboxes and oxygen tanks, reaching mAP@0.5 of 0.949 (precision 0.91, recall 0.86).
- Improved robustness to occlusion and low light through mosaic and low-light augmentation; ~21.5 ms inference per image on a T4 GPU.
- Implemented the end-to-end pipeline: data loading, training, evaluation and annotated prediction export.

ResumeSnap — Resume Analysis & Job Matching, *Python · Gemini API · OCR · Scikit-learn · Streamlit*
Live demo: https://piyushkarn-resumesnap.hf.space/

- Built an OCR and LLM extraction pipeline that converts unstructured resumes into structured records, with cosine-similarity ranking against job descriptions across a 5,000-record dataset.
- Sped up matching time from ~15 minutes of manual review to under 10 seconds per resume.

Achievements

1st place, Duality AI Track, *Hackwave 2025* 04/2025
VisionX, object detection for space-station safety.